THE HALTING PROBLEM OF THE FOUR TURING MACHINES CM16/26/51/52: CRITERIA, ERA ALGEBRA, CORRESPONDENCE FACTS, AND THE OPEN PROBLEM

1. The four machines and conventions

Machine	Transition table	Halting transition
cm16	`1RB0LE_0RC1RB_1LC1RD_1RF0LA_0RD1LE_1RB---`	F on 0
cm26	`1RB1RC_1LC0LE_1RD1RC_1LB0RA_---0LF_1RD1LF`	E on 0
cm51	`1RB1RA_1LC0RC_1RE0LD_0RE0LF_1RA---_1RB1LF`	E on 0
cm52	`1RB0LD_1LC1RD_0LE1LA_0RB1RD_---0LF_1RA1LF`	D on 0

Bi-infinite tape, initially all 0, head at cell 0, state A. The four halting transitions occupy different positions; criteria are established independently per machine (§2), followed by the common era algebra (§3), the verified correspondence facts (§4), and the per-machine reduction and open problem (§5–§6). All criterion- and routine-layer assertions are exhaustively verified by two checkers: `casecheck-halt.mjs` (204,800 configurations, 111,257 halting instances, zero counterexamples, entry enumeration ENUM-OK ×4) and `casecheck-island.mjs` (39 assertions, all passing).

2. Halt criteria (independent per machine)

Theorem 2.1 (cm16)**.** *cm16 halts if and only if at some time state D reads a 0 at cell w with $\tau(w{+}1) = 1$.*

Theorem 2.2 (cm26)**.** *cm26 halts if and only if at some time state B reads a 1 at cell q with $\tau(q{-}1) = 0$.*

Theorem 2.3 (cm51)**.** *cm51 halts if and only if at some time state C or D reads a 0 at cell w with $\tau(w{+}1) = 1$.*

Theorem 2.4 (cm52)**.** *cm52 halts if and only if at some time state C reads a 0 at cell c with $\tau(c{-}1) = 0$.*

Method (identical form for all four). Halting $\iff$ reaching the machine's halting state reading 0. Enumerate at transition level all in-edges of the halting state (as in Theorem 4.1 of the cm2 paper: trace the provenance of the cell values read), reducing "halting state reads 0" to the visit of a specific two-cell local pattern; then instantiate both directions over all configurations in the pattern window: configurations satisfying the pattern halt within finitely many steps, the others leave the window along their routine branches. Exhaustiveness via the checker: frozen skeleton (per machine, from deep scans) × working-triple boundaries × head classes, 204,800 cells, both directions, zero counterexamples [casecheck-halt]. □

3. Lap grammar and era algebra

Definition 3.1 (marker states and eras)**.** *A marker state is the (A, h, τ) at the end of a dance-family routine; the evolution between consecutive markers is one era. The blank-tape initial segment enters the first marker by direct simulation.*

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Lemma 3.2 (standard lap word)**.** *For each machine and each parameter $R \geq 2$, the (state, symbol) event word of one lap macro step is*

$$\underbrace{A0\,B0\,C1\,D1\,A1\,E0\,D0\,F0}_{8} \;\Big|\; \underbrace{B1^{R-2}}_{R-2}\;\underbrace{B0\,C0\,C0\,C1\,D1\,A1}_{6} \;\Big|\; \underbrace{E1^{R-1}}_{R-1}\;\underbrace{E0\,D1\,A0}_{3},$$

$8+(R-2)+6+(R-1)+3 = 2R+14$ *events in total (an event = one (state, symbol); the final $A0$ is the boundary event shared with the next lap, the lap proper being $2R+13$ transitions); net effect $(x, R, z) \mapsto (x+1,\ R-2,\ z+1)$, the working window shifting one cell right with the head in the same relative position.*

complete derivation for cm16; the other three machines analogously. Let the left seam be cells $g, g+1$ (values $0, 0$), the right reservoir 1^R occupying $g+2..g+R+1$, the right seam $g+R+2, g+R+3$, the deposit 1^z from $g+R+4$. Head A at g.

Segment 1 (8 steps): $A0$=`1RB` writes $1@g$, right; $B0$=`ORC` at $g+1$; $C1$=`1RD` at $g+2$; $D1$=`0LA` writes $0@g+3$, left; $A1$=`0LE` writes $0@g+2$, left; $E0$=`ORD` at $g+1$; $D0$=`1RF` writes $1@g+2$, right; $F0$=`1RB` writes $1@g+3$, right. Now $g+2, g+3 = 1, 1$, head B at $g+4$ (inside the reservoir). Between steps 7 and 8 the tape shows a transient island ($g+1, g+2, g+3 = 0, 1, 0$) with head distance ≤ 1.

Segment 2 ($R+4$ steps): $B1$=`1RB` keeps writing 1 moving right across $g+4..g+R+1$ ($R-2$ cells); $B0$ at $g+R+2$ (`ORC`); $C0\,C0$ = two applications of `1LC` writing 1 at $g+R+3$ then $g+R+2$ (moving left); $C1$ at the reservoir end $g+R+1$ (`1RD`); $D1$ writes $0@g+R+2$; $A1$ writes $0@g+R+1$, moving left into E at $g+R$.

Segment 3 ($R+2$ steps): $E1$=`1LE` keeps writing 1 moving left back across $g+R..g+2$ ($R-1$ cells); $E0$=`ORD` at $g+1$; $D1$=`0LA` writes $0@g+2$, ending at A on $g+1$ — the start of the next lap.

Net effect check: the final tape has g=1 (left block +1); new left seam $g+1, g+2$; reservoir remnant $g+3..g+R$ ($R-2$ cells); new right seam $g+R+1, g+R+2$; $g+R+3$=1 absorbed into the deposit (+1). That is $(x+1,\ R-2,\ z+1)$, with the next lap's head on the left cell of the new left seam — same relative position, window shifted by one. Step count $8+(R-2)+6+(R-1)+3 = 2R+14$.

Cross-machine remark: the word is *event-level* (a sequence of (state, symbol) pairs) — the move directions of the transitions may differ between machines (e.g. $D1$ is `0LA`, moving left, on cm16 but `0RA`, moving right, on cm26), yet the event sequence and net bookkeeping coincide; the per-machine geometry is verified by cell-diff on deep-scan anchors (about 780,000 laps in total) [casecheck-island]. □

Definition 3.3 (eras and resets)**.** *One era = the lap chain up to reservoir exhaustion + the drain routine + the reset. Reservoir parameters (L, M, R) evolve along the chain by Lemma 3.2; draining triggers a churn cascade (reseeding). Era key sets (key = (position class, neighborhood block count)) collected from deep scans: cm16/26/51 share 78 keys identically; cm52 has 89.*

4. The era system: data and structure

Definition 4.1 (reservoir configuration)**.** *The marker tape consists of: a left reservoir 1-block, a right reservoir 1-block and a deposit block, separated by 0^2 seams, with the frozen skeleton as tail (identical bit-for-bit across the four machines, fixed by deep scans); the working triple (x, R, z) records the working-block, right-reservoir and deposit lengths.*

Remark 4.2 (era-chain example (cm16, near 10^9 steps))**.** *The measured working-triple sequence of eras:*

$$(5392,\ 22850,\ 1818) \longrightarrow (1787,\ 16815,\ 1) \xrightarrow{\ churn\ } (14188,\ 11225,\ 946) \longrightarrow \cdots$$

The right reservoir decreases by steps of 2 along the lap chain until exhaustion, triggering the drain and the churn cascade (reseeding into a new reservoir layout); the chain truncations and reset branches form the era key set.

Remark 4.3 (cross-machine alignment data)**.** *The large-era* (L_0, R_0) *sequences of the four machines align over 562 measured pairs at constant offsets (*0 *or* +1*) with zero misalignment (deep-scan prefix, at the* 5×10^9*-step scale); the era key sets are 78 keys, identical key-by-key, for cm16/26/51 and 89 for cm52 (due to criterion and skeleton differences). These are observations (grade B), not used as theorems.*

5. Verified correspondence facts between the four machines

Remark 5.1. *The following are layered facts, not an isomorphism theorem (the four transition tables differ; no transition-level isomorphism exists):*

1. *the lap grammar and bookkeeping are the same on all four machines (Lemma 3.2, verified per machine);*
2. *the large-era* (L_0, R_0) *sequences align over 562 pairs at constant offsets (*0 *or* +1*) with zero misalignment — an observation on the deep-scan prefix [at the* 5×10^9*-step scale];*
3. *end configurations agree bit-for-bit only on the frozen skeleton; working triples differ by* ± 100 *(a correction of the earlier "bitwise identical" measurement);*
4. *the halt criteria have different sources (Theorems 2.1–2.4); non-halting assembly proceeds independently per machine.*

6. The island invariant and the danger pattern

Definition 6.1 (island)**.** *An island is the pattern* $(0^1, 1^1, 0^1)$ *(a single-cell* 1*-island). A danger event is an occurrence of this pattern with the* ***head far*** *(not co-located with the destroyer wave).*

Theorem 6.2 (island invariant)**.** *Within the verified domain (*2×10^7 *steps per machine, 892 island moments): every island occurrence is co-located with the destroyer wave (head distance* ≤ 1*); island occurrences are transient — within each lap islands exist in* two *segments (correcting the earlier "only the boundary* R*=1 segment"), with creator and destroyer the same passing wave and existence window shorter than one lap [casecheck-island, 39 assertions].*

Definition 6.3 (vocabulary, audit and guard)**.** *The* verified vocabulary *is the set of routine branches actually occurring and catalogued in deep scans and censuses (the standard lap word with its parameter branches, drain branches, churn reseed branches; the authoritative inventory is the era data and the classifier of* `quad.c`*).* Audit mode*: at every A-state landing, check whether the danger pattern (a head-far* $(0^1, 1^1, 0^1)$*) appears in the five-cell window (head neighborhood* ± 2*), stepping to a given depth.* Guard mode*: at every step, check directly whether the machine's criterion (the two-cell condition of Theorems 2.1–2.4) holds, stepping to a given depth.*

Theorem 6.4 (safety within the vocabulary)**.** *At every A-state landing covered by the verified routine vocabulary, the pattern preconditions of Theorems 2.1–2.4 do not occur (sprint-landing right gaps* ≥ 2*; island positions excluded by Theorem 6.2).*

7. Reduction and open problem

Theorem 7.1 (reduction, per machine)**.** *For each machine* $m \in \{16, 26, 51, 52\}$ *the following are equivalent:*

1. *m halts;*
2. *the criterion pattern of m (the two-cell condition of Theorems 2.1–2.4) holds at an* out-of-vocabulary *landing (a danger event).*

Machine verification: audit mode steps through five-cell checks at out-of-vocabulary landings (10^9 *steps* $\times 4$*) with zero danger occurrences; guard mode (*2×10^{11} *steps* $\times 4$*, stepwise halt-precondition checking) with zero triggers; bare-machine deep scan of cm16 to* 5×10^{11} *steps with zero halts.*

Proof. (2)⇒(1): the criterion pattern triggers the in-edge chain of Theorems 2.1–2.4; the machine halts. (1)⇒(2): suppose a halt; then some criterion pattern is visited; by Theorem 6.4 (vocabulary safety) the visit occurs at an A-state landing outside the vocabulary coverage; the enumeration completeness of out-of-vocabulary landings is carried by the census/era key coverage [A^-]. □

Conjecture 7.2 (five-cell avoidance)**.** *No danger event ever occurs in the runs of the four machines (formally: clause (2) of Theorem 7.1 is false for all time).*

Refutation record: the local-abstraction route (key graphs) is dead — in the radius-ladder experiment for key-set successor closure, $|K^*|$ grows by about $\times 3.1$ per added radius (cm16: $99 \to 294 \to 939 \to 2863 \to 8879$ for $r = 2..6$; two reruns reproduce digit-for-digit), with danger-key hits growing in step, so no finite fixed point exists for fixed-radius local abstraction [kclosure]. Research field: piecewise-affine trajectory avoidance of the same shape as the cm2/M1 residuals (classification by analogy; per-machine formal restatement not performed).

Reproduction

`node casecheck-halt.mjs && node casecheck-island.mjs && node kclosure.mjs`; `quad6.exe <code> audit 1000000000`; `quad6.exe <code> guard 200000000000`. All one-line commands.